

Giulia Rocco

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Researcher | Multimodal Neuroimaging & Signal Processing Expert

Experienced in multimodal data acquisition and advanced signal processing, specializing in high-dimensional time-series modeling and functional brain mapping. Passionate about bridging neuroscience and technology. Quick learner, proven ability to design and execute complex experimental research studies across interdisciplinary teams.

WORK EXPERIENCE

I3S Lab and Centre Hospitalier Universitaire de Nice • Nice, France 01/2020 – 11/2023 Graduate Research Assistant

- Conducted multimodal data collection and analysis (fMRI, fNIRS, EEG) to investigate cerebellar activation, resulting in 6 published findings.
- Designed and validated non-invasive novel experimental frameworks for studying cerebellum, earning awards for breakthroughs in cerebellar neuroscience.
- Led cross-institutional project management, coordinating 50+ subject recruitment, 20K \$ budget oversight, and data integrity, while also successfully obtaining two equipment loans from industry to facilitate experiments.

MultifunkIM Lab, Concordia University • Montreal, Canada 03/2022 – 12/2022 Visiting Researcher

- Pioneered a computational framework for cerebellar fNIRS image reconstruction, effectively solving forward and inverse problems.
- Led multimodal data collection projects involving 30 participants and 300+ recordings.

NIRx Medical Technologies • Berlin, Germany 06/2022 – 07/2022 PhD Intern

- Developed a customer-facing document detailing the value proposition of simultaneous fNIRS/fMRI, including hardware and software summaries, and drafted a video plan for data analysis software promotion.

Université Côte d'Azur • Nice, France 09/2020 – 12/2020 Teaching Assistant

- Taught Algorithms & Programming with R to 30+ students, delivering practical sessions and project-based assignments.

B3Lab and PHEEL Laboratories, Politecnico di Milano • Milan, Italy 01/2019 – 11/2019 Research Intern

- Analyzed biomedical signals (EEG, ECG, PPG, EDA, eye tracking) for emotion and stress detection, developing novel indices for biomedical and biomarketing applications.
- Contributed to multidisciplinary VR therapy research for neurodevelopmental disorders, managing physiological monitoring and data analysis to drive insights.
- Conducted a comparative analysis of two popular wearable devices and a clinical-grade ECG, revealing significant variations in inter-beat-interval accuracy.

Audio Engineering Lab, KU Leuven • Leuven, Belgium

10/2017 – 03/2018

Visiting Scholar

- Design and implementation of a novel language-independent speech intelligibility test for hearing screening, resulting in 3 publications.
- Characterized speech stimuli intelligibility and psychometric functions with different noise realizations.

Lutech • Milan, Italy

10/2015 – 12/2015

Support Engineer

- Resolved technical issues and provided immediate support to operating room staff during the EHR framework deployment, minimizing disruptions and maximizing efficiency.

LAFAS Lab, Università di Milano • Milan, Italy

03/2015 – 07/2015

Visiting Student

- Validated and optimized machine learning algorithms for CT image segmentation, achieving a correlation of over 0.9 with manual methods and reducing processing time by 53%.

EDUCATION

PhD (MSCA-COFUND) in Automatics, Signal and Image processing

Université Côte D'Azur • Nice, France

Master Thesis in Biomedical Audio Engineering

KU Leuven • Leuven, Belgium

Erasmus+ mobility in Engineering

KTH Royal Institute of Technology • Stockholm, Sweden

Master's Degree in Biomedical engineering – Electronic Technologies

Politecnico di Milano • Milan, Italy • GPA: 110/110

Bachelor's Degree in Biomedical Engineering

Politecnico di Milano • Milan, Italy • GPA: 102/110

AWARDS & SCHOLARSHIPS

Grant for Pilot Projects

Quebec Bio-Imaging Network (20K\$)

10/2022

Excellence Prize

Université Côte D'Azur

12/2021

Winner Emerging Talent

MSCA Falling Walls Lab – Listed among the 75 International Emerging Talents

11/2021

Best Poster Award

4th Congress of the Physiology and Integrative Biology French Society

07/2021

Public Award, 3-minutes Thesis contest

18th IEEE International Symposium on Biomedical Imaging

04/2021

Doctoral Scholarship

Marie Skłodowska-Curie Actions COFUND (15 selected over 500 applicants)

01/2020

Merit Scholarship

Politecnico di Milano

09/2016

SKILLS

Programming: Matlab, Python, R, Git, C++

Quantitative: Time-series analysis, Regression analysis, Machine learning, Hypothesis testing, Inverse problems, Monte Carlo simulations

Language: English, Italian, French

PUBLICATIONS

Characterization of the Intelligibility of Vowel-Consonant-Vowel (VCV) Recordings in Five Languages for Application in Speech-in-Noise Screening in Multilingual Settings

Applied Sciences

Enhancing Cerebellar fNIRS/fMRI via Tailored Pipelines

Biennial meeting of the Society of Functional Near-Infrared Spectroscopy (SfNIRS 2024)

Densifying Optodes Montage to Enhance Cerebellar fNIRS

Biennial meeting of the Society of Functional Near-Infrared Spectroscopy (SfNIRS 2022)

When fNIRS meets fMRI to complement cerebellar exploration

IEEE International Symposium on Biomedical Imaging (ISBI 2022)

A chiral fNIRS spotlight on cerebellar activation in a finger tapping task

43rd Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC)

Single-Trial Detection of Event-Related Potentials with Integral Shape Averaging : An Application to the Elusive N400

43rd Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC)

Asymmetries in Cerebellar Activation during Finger Movements : A Functional Near-Infrared Spectroscopy Study

Acta Physiologica : 4th Congress of Physiology and Integrative Biology, 88th Congress of French Physiological Society, Faculté de Médecine, Nice, France, 2-4 September, 2021

Exploring the cerebellum with functional near-infrared imaging : a preliminary study

IEEE 18th International Symposium on Biomedical Imaging (ISBI 2021)

P300 event-related potentials classification from EEG data through interval feature extraction and recurrent neural networks

SophIA Summit 2020

An automated speech-in-noise test for remote testing : Development and preliminary evaluation

American journal of audiology

Characterization of eye gaze and pupil diameter measurements from remote and mobile eye-tracking devices

XV Mediterranean Conference on Medical and Biological Engineering and Computing-MEDICON 2019 : Proceedings of MEDICON 2019 , September 26-28 , 2019 , Coimbra, Portugal

Development and preliminary evaluation of a novel adaptive staircase procedure for automated speech-in-noise testing

2019 41st Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC)

Heart rate variability from wearables : a comparative analysis among standard ECG, a smart shirt and a wristband

pHealth 2019

Validation of an automatic hard tissue segmentation algorithm for cone beam CT data

37th IEEE Annual conference on Engineering in Medicine and Biology